

BIOSTATR: Applied Biostatistics with R

Full Program, September-October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Applied Biostatistics with R

Format: Online, cohort-based

Duration: 3 weeks, 9 sessions, 2 hours per session, 18 contact hours

Dates: 07/09/2026 – 25/09/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00)

Time: Morocco time

Instructor: Imad EL BADISY, PhD

Contact: elbadisyimad@gmail.com | elbadisyimad.com

1.1 Course Description

BIOSTATR is the base course of the MHDSR pathway: a practical and rigorous introduction to applied biostatistics for health research using R. It takes participants from a clean R and data-manipulation workflow through descriptive analysis, inference, linear and logistic regression, survival analysis and flexible survival models, missing-data handling with multiple imputation, and meta-analysis, all with reproducible Quarto reporting.

Meta-analysis is taught here as a biostatistical method, not as a review-methodology course. The course emphasizes estimation, interpretation, reproducibility, and the appropriate use of statistical methods in clinical research.

1.2 Target Audience

Clinicians, biomedical researchers, PhD students, public health professionals, and analysts. No advanced statistics background required.

1.3 Prerequisites

Basic computer literacy. No prior R experience required; R and RStudio setup is covered in Session 1.

1.4 Learning Objectives

By the end of BIOSTATR, participants will be able to:

1. organize a reproducible analysis with R, Quarto, and version control;
2. import, clean, reshape, and quality-check health datasets;
3. summarize and visualize health data appropriately, including stratified summaries;
4. interpret confidence intervals, p-values, effect sizes, power, and sample size;
5. compare groups with suitable tests for quantitative and qualitative outcomes;
6. fit and interpret linear and logistic regression, including confounding and interactions;
7. analyze time-to-event data with Kaplan-Meier, the log-rank test, and the Cox model;
8. model non-linear and time-dependent effects with flexible survival models;
9. handle missing data with multiple imputation and Rubin's rules;
10. carry out a meta-analysis: effect sizes, fixed- and random-effects models, heterogeneity, and publication bias.

2 Detailed Schedule

Table 1: BIOSTATR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	07/09/2026 (Mon)	18h00- 20h00	Introduction to R	Environment, projects, scripts, objects and types, import/export, good practices, reproducible reports
2	1	09/09/2026 (Wed)	18h00- 20h00	Data manipulation	Filtering, transformation, joins, pivots, dates, strings, recoding, quality control
3	1	11/09/2026 (Fri)	18h00- 20h00	Descriptive statistics and visualization	Descriptive statistics, distributions, summary tables, Table 1, graphics, stratification
4	2	14/09/2026 (Mon)	18h00- 20h00	Statistical inference and hypothesis testing	Estimation, confidence intervals, tests, effect sizes, power, sample size, quantitative and qualitative comparisons
5	2	16/09/2026 (Wed)	18h00- 20h00	Linear and logistic regression	Adjusted coefficients, odds ratios, confounding, interactions, diagnostics, predictions, ROC curve
6	2	18/09/2026 (Fri)	18h00- 20h00	Survival analysis	Censoring, Kaplan-Meier, log-rank test, Cox model, hazard ratios, proportional-hazards assumption
7	3	21/09/2026 (Mon)	18h00- 20h00	Flexible survival models	Non-linearity, splines, restricted cubic splines, flexible parametric models, time-dependent effects, survival predictions

No.	Week	Date	Time	Topic	Main Content
8	3	23/09/2026 (Wed)	18h00- 20h00	Missing data and multiple imputation	MCAR/MAR/MNAR, complete-case analysis, single and multiple imputation, imputation model, Rubin's rules, sensitivity analyses
9	3	25/09/2026 (Fri)	18h00- 20h00	Meta-analysis	Effect sizes, fixed- and random-effects models, forest plot, heterogeneity, meta-regression, publication bias

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

The course ends with a short reproducible Quarto report or mini-project, presented and discussed in the final session.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required; setup covered in Session 1.

Attendance: Full attendance at all 9 sessions is recommended given the cumulative structure.